



In addition to the part-I (General Handout for all courses appended to the timetable), this portion gives further specific details regarding the course.

Course No. :	CHE G554
Course Title :	COMPUTATIONAL FLUID DYNAMICS
Instructor-in-charge:	SAIKAT BHATTACHARJEE

1. **Course Description:** Philosophy of Computational Fluid Dynamics (CFD); governing equations of fluid dynamics; mathematical behaviour of partial differential equations and their impact on CFD; finite volume and finite difference discretization, the incompressible Navier-Stokes equations. Iterative methods, simulation procedures for sample chemical engineering problems using FLUENT.
2. **Scope and Objective:** This course aims to apply numerical methods and transport phenomena to understand and use CFD as a scientific tool. *The course is research-oriented. A student is expected to do an extensive guided reading of CFD applications and practice with building CFD models using ANSYS and COMSOL. The students will review and present research papers using CFD as investigative tools in class.*

3. Course Learning Outcomes

By the end of this course, students will be able to:

1. Explain the philosophy, scope, and applications of Computational Fluid Dynamics (CFD) in research and engineering design contexts.
2. Derive and interpret the governing equations of fluid dynamics and explain their physical significance in various flow conditions.
3. Classify and analyze partial differential equations relevant to CFD and evaluate their physical and computational implications.
4. Apply finite difference and finite volume methods for the discretization of governing equations and assess stability, accuracy, and convergence of numerical schemes.
5. Demonstrate the concepts of grid generation, transformation, and meshing techniques for CFD simulations.
6. Implement various CFD numerical techniques (e.g., Lax-Wendroff, MacCormack, ADI, Pressure Correction) for solving fluid flow problems.
7. Simulate incompressible Couette flow and other benchmark problems using appropriate implicit methods and pressure correction algorithms.
8. Use commercial CFD software (e.g., ANSYS Fluent, COMSOL) to model and analyze different types of fluid flow, heat transfer, species transport, combustion, multiphase flows, and moving mesh problems.
9. Critically review and present research literature applying CFD to chemical engineering and related fields.
10. Design and execute CFD-based projects, interpret simulation results, and present findings effectively in written and oral formats.

4. Text Book:

- TB1: J.D.Anderson, Computational Fluid Dynamics, Mcgraw Hill Inc., 1995, 6th Edition.
- TB2: An introduction to computational fluid dynamics: The finite volume method. H.K.Versteeg and W.Malajasekra, Longman





4. Reference Books:

- R1: Pozrikidis, C, Introduction to theoretical and computational fluid dynamics, Oxford university press, 1997
- R2: Lohner, Rainald, Applied computational fluid dynamics techniques: an introduction based on finite element methods, John Wiley & sons ltd. 2008
- R3: Fletcher, C.A.J, Computational techniques for fluid dynamics: specific techniques for different flow categories, Springer-Verlag 1991, second Edition

5. Course Plan:

Module	Lecture session/Tutorial Session (content)	Reference (TB)	Learning Outcome
1. CFD: definition and philosophy	L1-2: Computational Fluid Dynamics: Why? Computational Fluid Dynamics as a Research Tool, Computational Fluid Dynamics as a Design Tool, The Impact of Computational Fluid Dynamics, Applications.	1.1-1.5	• Explain the role of CFD as a research and design tool across engineering applications. • 2. Identify and discuss real-world examples where CFD has impacted process design and analysis.
2. CFD governing equations	L3: Introduction to the models of the fluid flow, Finite Control Volume, Infinitesimal Fluid Element, L4: The Substantial Derivative, The Divergence of the Velocity: Its Physical Meaning, L5: The Continuity Equation, The Momentum Equation, L6: The Energy Equation, Summary of the Governing Equations for Fluid Dynamics.	2.1-2 2.3-4 2.5 2.4	• Derive the continuity, momentum, and energy equations for fluid flow. • Interpret the physical meaning of each governing equation in the context of chemical engineering applications.
3. Mathematical behaviour of PDEs.	L7-8: Introduction, Classification of Quasi-Linear Partial Differential Equations, General Method of Determining the Classification of Partial Differential Equations: The Eigenvalue Method, L9-10 General Behavior of the Different Classes of Partial Differential Equations: Impact on Physical and Computational Fluid Dynamics, Hyperbolic Equations, Parabolic Equations, Elliptic Equations, The Supersonic Blunt Body	3.2-3.3 3.4-3.6	• Classify partial differential equations (PDEs) into hyperbolic, parabolic, and elliptic types using analytical methods. • Evaluate the implications of PDE classification on the physical behaviour of fluid flow and CFD solution approaches.





4. Discretization	L11-13 Introduction, Introduction to Finite Differences, Difference Equations, Explicit and Implicit Approaches: Definitions and Contrasts, L14 Errors and an Analysis of Stability, Stability Analysis: A Broader Perspective	4.1-4.4 4.5	• Apply finite difference and finite volume methods to discretize fluid flow equations. • Assess the stability, accuracy, and convergence of numerical schemes through error analysis
5. Grid transformations	L15-16 General Transformation of the Equations, Metrics and Jacobians. L17 Form of the Governing Equations Particularly Suited for CFD Revisited L18-19 Grid types and developments	5.1-3 5.4 5.5-10	• Explain the role of coordinate transformation, metrics, and Jacobians in CFD. • Compare different grid generation methods and their suitability for various CFD problems.
6. CFD Techniques & Algorithms	L20-21 The Lax-Wendroff Technique L22-23 MacCormack's Technique L24-25 Conservation Form and Space Marching. L26-27 The Relaxation Technique and Its Use with Low-Speed Inviscid Flow, Aspects of Numerical Dissipation and Dispersion; L28-30 Artificial Viscosity, The Alternating-Direction-Implicit (ADI) Technique. L31-32 The Pressure Correction Technique: Application, Computer Graphic Techniques Used in CFD	6.1-2 6.3 6.4 6.5-6 6.7 6.8	• Implement common CFD numerical techniques such as Lax-Wendroff, MacCormack, ADI, and pressure correction methods. • Evaluate the effects of numerical dissipation, dispersion, and artificial viscosity on CFD results.
7. Modelling incompressible Couette Flow	L33 The Physical Problem and Its Exact Analytical Solution The Numerical Approach: Implicit Crank-Nicholson Technique, The Pressure Correction Method	9.1-4	• Formulate and solve incompressible Couette flow problems using implicit and pressure correction methods. • Validate CFD solutions by comparing with exact analytical solutions for Couette flow.
8. Case Studies & Simulations using CFD	L34-42 Case studies using using ANYS FLUENT, COMSOL	Class material	• Perform CFD simulations for industrial case studies involving





Software	1. Modeling flow through mixing elbow 2. Modeling 2D Laminar Flow Past a Cylinder 3. Modelling incompressible Couette Flow 4. Modeling of turbulent flow in pipe using ANSYS 5. Modeling of fluid flow and heat transfer using ANSYS 6. Modeling of reaction and diffusion using COMSOL and ANSYS 7. Modeling Species Transport and Gaseous Combustion 8. Modeling Radiation and Natural Convection 9. Modeling Cavitation 10. Modelling fluidization using multi-phase Eulerian model 11. Modeling of moving parts using Sliding Meshes		single-phase, multiphase, heat transfer, and reacting flows. • Interpret and present simulation results effectively, linking them to theoretical concepts and engineering practice.
----------	--	--	--

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	25%		CB
Final Project	2 to 3 weeks	15%	To be announced in the class	Take home
Case Studies /Simulations/Assignments/lecture tests (4)	continuous	20%		Take Home
Comprehensive Examination	180 Min.	35%		CB&OB

6. Make-up Policy:

Make-up for Mid Semester and Comprehensive exams will be granted only in case of illness justified with a medical certificate from BITS CMO. An only medical prescription will NOT be considered. There is no make-up for any class evaluative (Class testing/ class project). Any make-up request without prior permission will not even be considered.

7. Chamber Consultation Hour:

To be announced in the class.

8. Notices: all announcements will be made on Nalanda only.

Instructor-In-Charge
CHE G554

